

1. Input

Section, materials and resistance basis exactly as analysed.

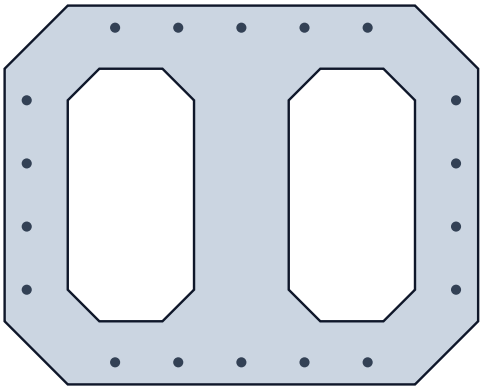
Identity

Project	PM-advanced (7) 2D reference case	Section	PM-advanced (7) 2D — 1500x1200 .
Calculation profile	KDS 2024 — Stress-strain integration	Mechanics	Stress-strain integration
Resistance profile	KDS 2024 current set · resistance: K.	Profile edition	2024 partial-revision framework (effective)
Verification status	draft	Concrete model basis	Code-default
Method id	kds-2024-current-set-global-strength.	Units	mm · N · MPa; forces reported in kN,.
Sign convention	Compression positive; $M_x = \Sigma F \cdot (y - y_c)$.		

Section

Cross-section (mm, about the net concrete centroid)

18 bars



Section properties

Gross concrete area A_g	1,120,000 mm ²
Reinforcement area A_s	14,476 mm ²
Reinforcement ratio ρ	1.293 %
Bar count	18
Net centroid	(0.0, 0.0) mm
Bounding box	1,500.0 × 1,200.0 mm
Solids / holes	1 / 2
Perimeter	9,262.7 mm

Materials

Name	KDS C30 (workbook Input)	Standard	KDS
f _{ck}	30.0 MPa	Model	kds-parabolic
ε _{cu}	0.00330	ε _{c0}	0.00200
α	0.8500	E _c	28,417 MPa
Tension	ignored	Name	SD400 (workbook Input)
Standard	KDS	f _y	400.0 MPa
E _s	200,000 MPa	Model	elastic-perfectly-plastic
ε _y	0.002000	ε _u	not declared
Grades used	1		

Design resistance basis

Format	Global resultant strength-reduction f.	Transverse reinforcement	other
φ compression (ties)	0.650	φ compression (spiral)	0.700
φ tension	0.850	ε _{t,limit}	0.00500
Transition rule	fixed-or-yield-multiple	Axial cap	0.800 × factored compression pole

Sampling and numerical evidence

Sampling mode	fixed	Active Design points	972
Active Design directions	36	Active Design stations	27
Active Nominal points	972	Active Nominal directions	36
Active Nominal stations	27	Direction tolerance	— %
Worst interpolation error	— %	Tolerance reached	yes
Concrete discretization	864 cells / 5196 points	Strain domain	concrete-pivot-ultimate

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is integrated from the material stress-strain law over a verified triangle/quadrature mesh of the section.

Overview interaction diagrams cut the active fixed 27-station, 10° direction surface. An arbitrary exact direction is calculated only when it is

requested.

Warning: Design profile status is draft; results are for review, not release.

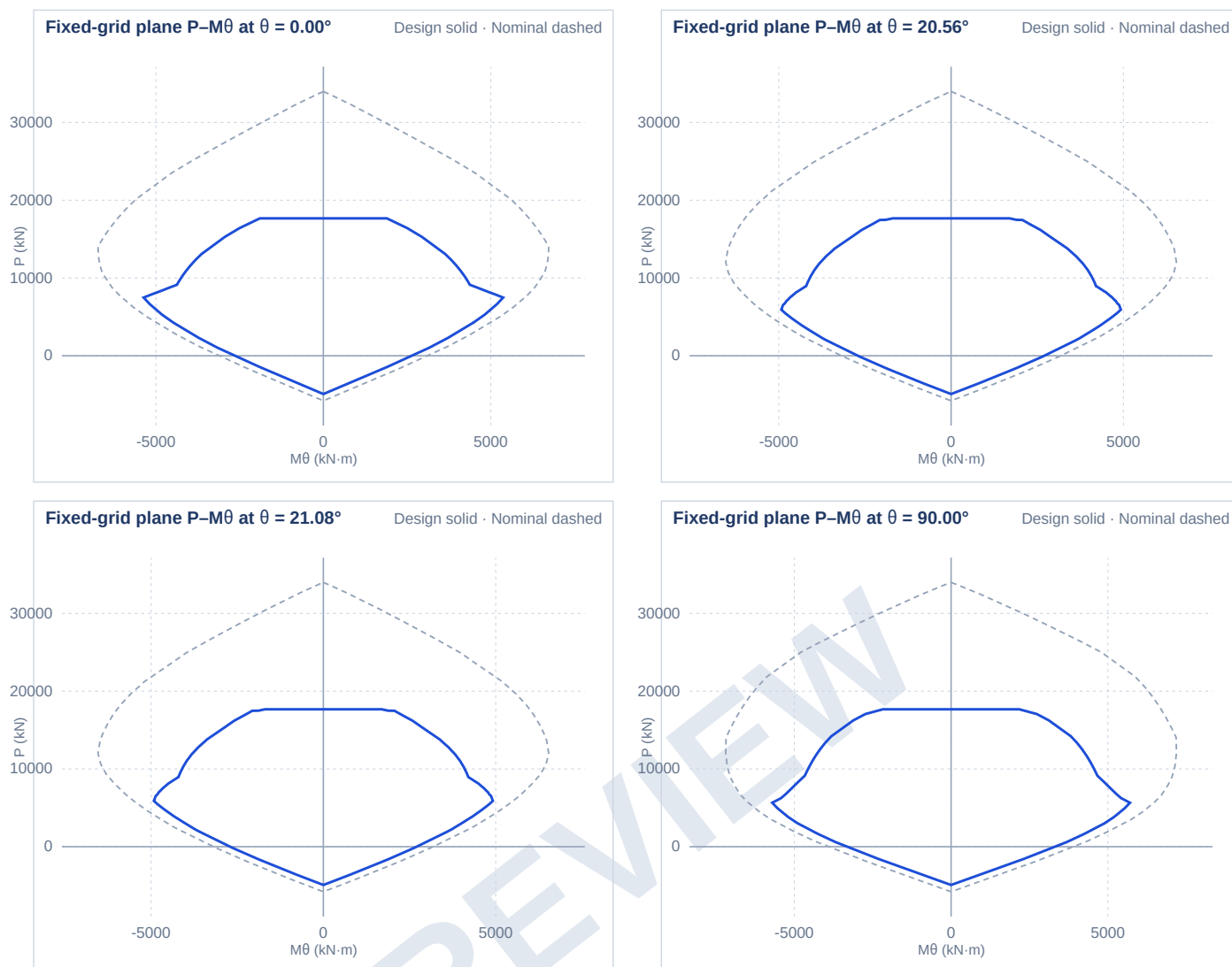
Reinforcement schedule

#	Bar id	Ø (mm)	X (mm)	Y (mm)	As (mm ²)	Steel
1	1	32.0	-400.0	530.0	804.2	SD400 (workbook Input)
2	2	32.0	-200.0	530.0	804.2	SD400 (workbook Input)
3	3	32.0	0.0	530.0	804.2	SD400 (workbook Input)
4	4	32.0	200.0	530.0	804.2	SD400 (workbook Input)
5	5	32.0	400.0	530.0	804.2	SD400 (workbook Input)
6	6	32.0	-400.0	-530.0	804.2	SD400 (workbook Input)
7	7	32.0	-200.0	-530.0	804.2	SD400 (workbook Input)
8	8	32.0	0.0	-530.0	804.2	SD400 (workbook Input)
9	9	32.0	200.0	-530.0	804.2	SD400 (workbook Input)
10	10	32.0	400.0	-530.0	804.2	SD400 (workbook Input)
11	11	32.0	-680.0	300.0	804.2	SD400 (workbook Input)
12	12	32.0	-680.0	100.0	804.2	SD400 (workbook Input)
13	13	32.0	-680.0	-100.0	804.2	SD400 (workbook Input)
14	14	32.0	-680.0	-300.0	804.2	SD400 (workbook Input)
15	15	32.0	680.0	300.0	804.2	SD400 (workbook Input)
16	16	32.0	680.0	100.0	804.2	SD400 (workbook Input)
17	17	32.0	680.0	-100.0	804.2	SD400 (workbook Input)
18	18	32.0	680.0	-300.0	804.2	SD400 (workbook Input)

2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 17670.3 kN; it appears as the flat top of each diagram.



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θL (°)	φ	Class	UR	Verdict
1	ULS-1 workbook Newton d.	24942.9	3714.2	1431.8	21.1	0.650	compression-co.	1.539	INADEQUA.
2	ULS-2 same axial, half mo.	24942.9	1857.1	715.9	21.1	0.650	compression-co.	1.412	INADEQUA.
3	ULS-3 low axial, high mom.	5000.0	4000.0	1500.0	20.6	0.850	transition	0.877	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

4.1 ULS-1 workbook Newton demand (on the 15 deg surface)

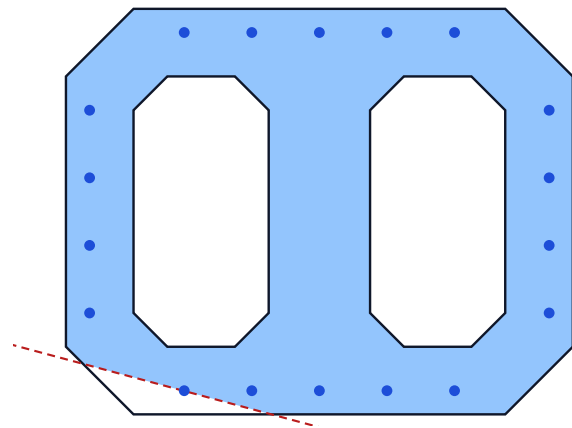
$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 3714.2 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1431.8 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

INADEQUATE · utilization 1.5394 · fixed-P diagnostic —

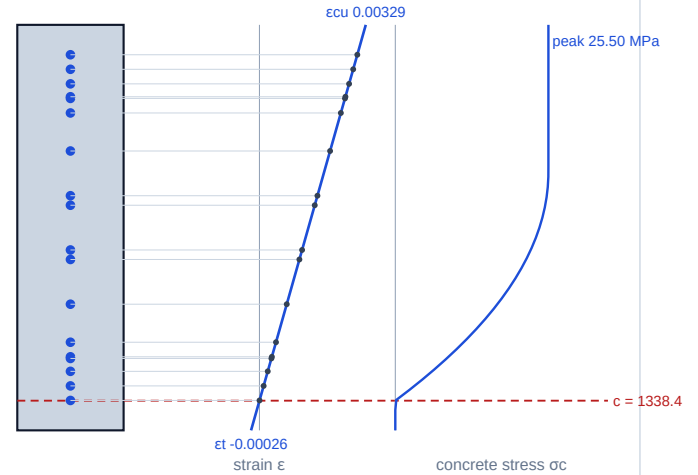
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta = 1444$

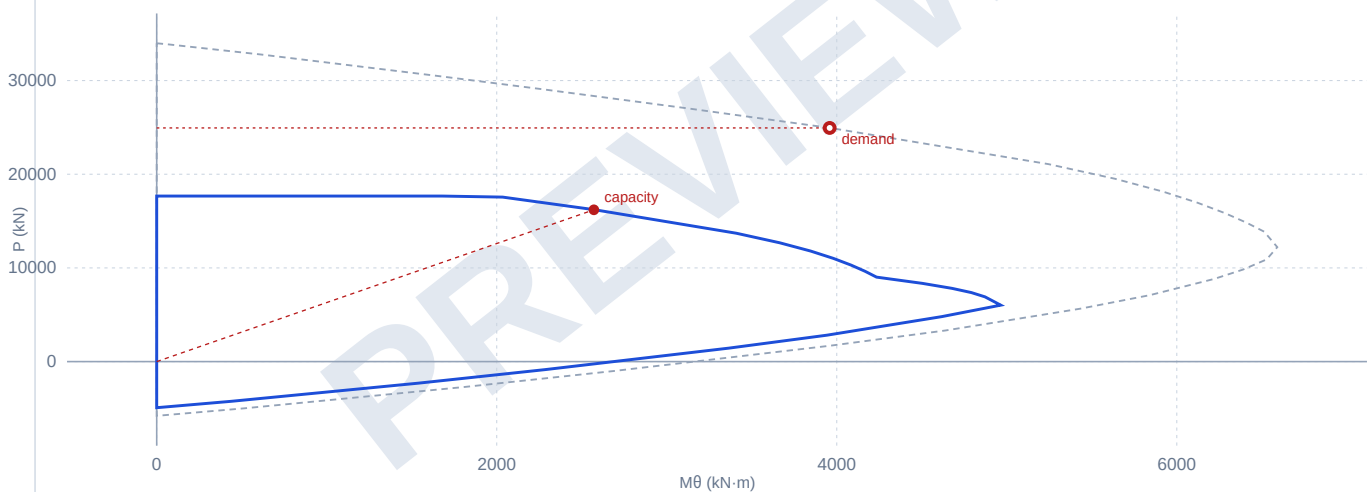


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P-Mβ at $\beta_{eq} = 15.019^\circ$, with the load point

solid = fixed Design · dashed = fixed Nominal · points are projected on β_{eq}



Concrete and resultants

Concrete force	21,577.4 kN	Concrete Mx	2,900.4 kN·m
Concrete My	1,122.3 kN·m	Steel force (gross)	3,629.5 kN
Displaced concrete	-264.0 kN	Steel net	3,365.5 kN
Response P	24,942.9 kN	Response Mx	3,714.2 kN·m
Response My	1,431.8 kN·m	Residual $ \Delta P $	1.01e-4 N
Residual $ \Delta M_x $	3.15e-3 N·mm	Residual $ \Delta M_y $	4.99e-3 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σ_s (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.8	2.520e-3	400.00	321.70	—
2	-200.0	530.0	804.2	262.0	2.648e-3	400.00	321.70	—
3	0.0	530.0	804.2	210.1	2.775e-3	400.00	321.70	—
4	200.0	530.0	804.2	158.3	2.903e-3	400.00	321.70	—

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.5	3.030e-3	400.00	321.70	—
6	-400.0	-530.0	804.2	1337.6	1.971e-6	0.39	0.32	—
7	-200.0	-530.0	804.2	1285.8	1.295e-4	25.89	20.82	—
8	0.0	-530.0	804.2	1233.9	2.570e-4	51.39	41.33	—
9	200.0	-530.0	804.2	1182.1	3.845e-4	76.89	61.84	—
10	400.0	-530.0	804.2	1130.3	5.120e-4	102.39	82.35	—
11	-680.0	300.0	804.2	608.5	1.795e-3	359.09	288.80	—
12	-680.0	100.0	804.2	801.7	1.320e-3	264.06	212.37	—
13	-680.0	-100.0	804.2	994.8	8.451e-4	169.02	135.94	—
14	-680.0	-300.0	804.2	1188.0	3.699e-4	73.99	59.50	—
15	680.0	300.0	804.2	256.1	2.662e-3	400.00	321.70	—
16	680.0	100.0	804.2	449.2	2.187e-3	400.00	321.70	—
17	680.0	-100.0	804.2	642.4	1.712e-3	342.42	275.39	—
18	680.0	-300.0	804.2	835.6	1.237e-3	247.38	198.96	—

Solver and admissibility evidence

Strain plane ϵ_0	1.5162e-3	Curvature k_x	2.3759e-6
Curvature k_y	6.3748e-7	Neutral-axis line	164.9806°
Compression depth c	1,338.387 mm	Iterations	7
Relative residual	2.576e-12	Max concrete compression	0.003292
Concrete limit	0.003300	Max steel tension	0.000000
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	1.5394		

4.2 ULS-2 same axial, half moment (inside)

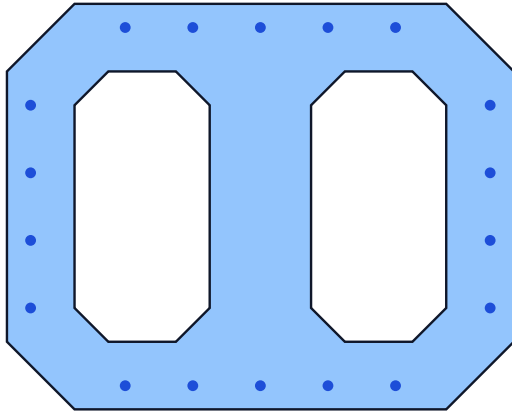
$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 1857.1 \text{ kN}\cdot\text{m}$ · $M_{uy} = 715.9 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

INADEQUATE · utilization 1.4116 · fixed-P diagnostic —

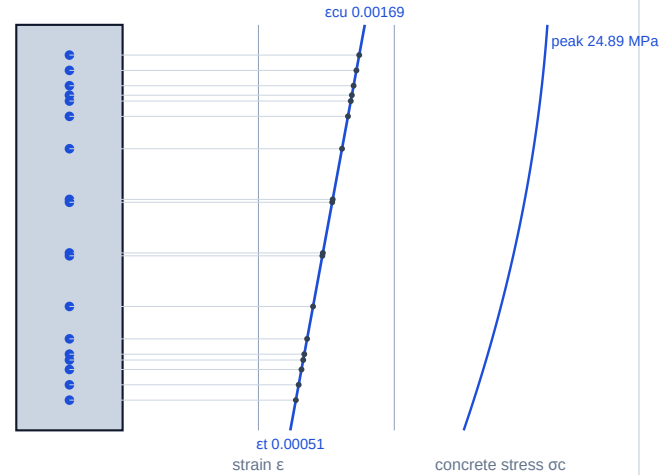
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta=1456$

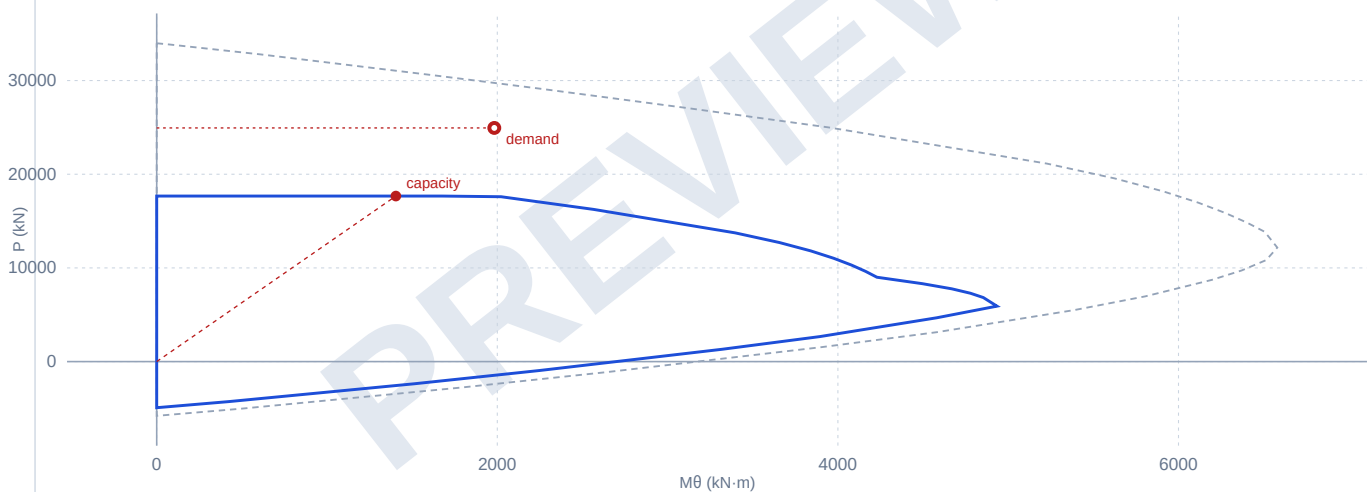


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P-Mβ at $\beta_{eq} = 15.968^\circ$, with the load point

solid = fixed Design · dashed = fixed Nominal · points are projected on β_{eq} $c = 2079.0$



Concrete and resultants

Concrete force	22,045.7 kN	Concrete Mx	1,476.8 kN·m
Concrete My	563.3 kN·m	Steel force (gross)	3,180.1 kN
Displaced concrete	-282.9 kN	Steel net	2,897.2 kN
Response P	24,942.9 kN	Response Mx	1,857.1 kN·m
Response My	715.9 kN·m	Residual ΔP	1.43e-5 N
Residual ΔMx	5.19e-3 N·mm	Residual ΔMy	1.92e-3 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	328.6	1.423e-3	284.64	228.92	—
2	-200.0	530.0	804.2	273.6	1.468e-3	293.59	236.12	—
3	0.0	530.0	804.2	218.6	1.513e-3	302.54	243.32	—
4	200.0	530.0	804.2	163.6	1.557e-3	311.49	250.51	—

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	108.6	1.602e-3	320.43	257.71	—
6	-400.0	-530.0	804.2	1347.7	5.946e-4	118.92	95.64	—
7	-200.0	-530.0	804.2	1292.7	6.393e-4	127.87	102.84	—
8	0.0	-530.0	804.2	1237.7	6.841e-4	136.81	110.03	—
9	200.0	-530.0	804.2	1182.7	7.288e-4	145.76	117.23	—
10	400.0	-530.0	804.2	1127.7	7.735e-4	154.71	124.42	—
11	-680.0	300.0	804.2	626.8	1.181e-3	236.16	189.93	—
12	-680.0	100.0	804.2	819.1	1.024e-3	204.89	164.78	—
13	-680.0	-100.0	804.2	1011.4	8.681e-4	173.62	139.63	—
14	-680.0	-300.0	804.2	1203.7	7.118e-4	142.35	114.49	—
15	680.0	300.0	804.2	252.7	1.485e-3	297.00	238.86	—
16	680.0	100.0	804.2	444.9	1.329e-3	265.73	213.71	—
17	680.0	-100.0	804.2	637.2	1.172e-3	234.46	188.57	—
18	680.0	-300.0	804.2	829.5	1.016e-3	203.19	163.42	—

Solver and admissibility evidence

Strain plane ϵ_0	1.0984e-3	Curvature k_x	7.8173e-7
Curvature k_y	2.2369e-7	Neutral-axis line	164.0318°
Compression depth c	2,079.010 mm	Iterations	5
Relative residual	3.635e-13	Max concrete compression	0.001690
Concrete limit	0.003300	Max steel tension	0.000000
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	1.4116		

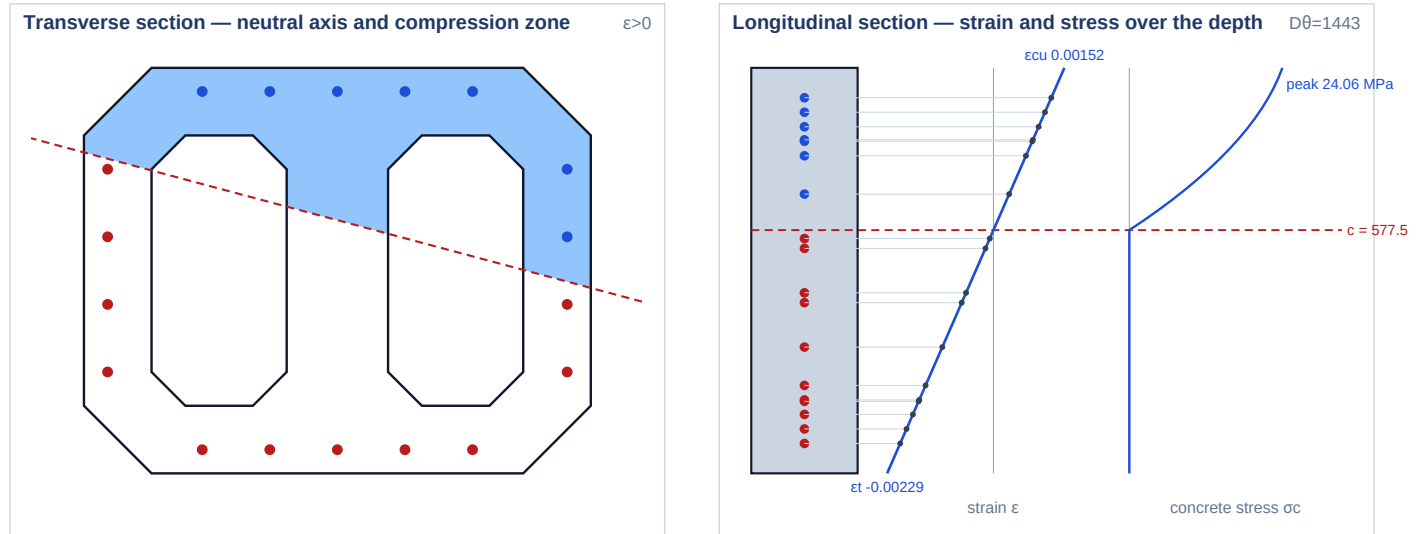
4.3 ULS-3 low axial, high moment

$P_u = 5000.0 \text{ kN}$ · $M_{ux} = 4000.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1500.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 20.56^\circ$

ADEQUATE · utilization **0.8774** · fixed-P diagnostic **0.9138**

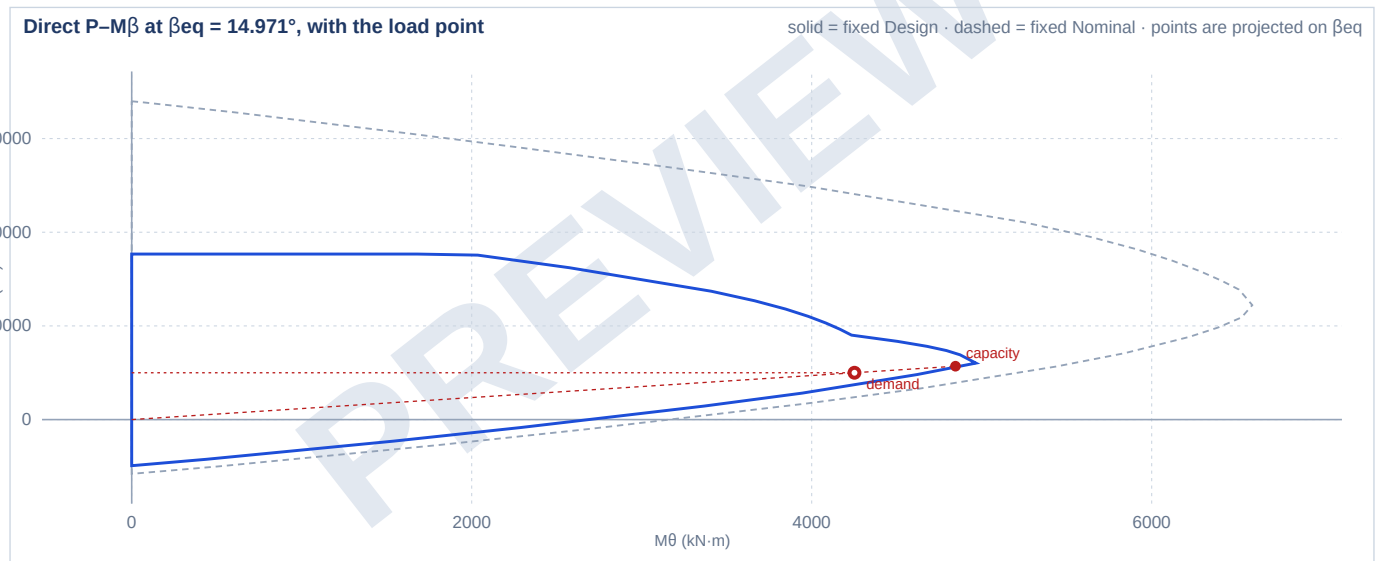
Converged preview equilibrium inside the material domain.

Section state



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction



Concrete and resultants

Concrete force	6,195.6 kN	Concrete Mx	2,728.2 kN·m
Concrete My	1,023.3 kN·m	Steel force (gross)	-1,101.1 kN
Displaced concrete	-94.5 kN	Steel net	-1,195.6 kN
Response P	5,000.0 kN	Response Mx	4,000.0 kN·m
Response My	1,500.0 kN·m	Residual ΔP	5.59e-9 N
Residual ΔMx	0.00e+0 N·mm	Residual ΔMy	9.54e-7 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.0	6.983e-4	139.66	112.33	—
2	-200.0	530.0	804.2	261.4	8.347e-4	166.94	134.26	—
3	0.0	530.0	804.2	209.7	9.711e-4	194.22	156.20	—
4	200.0	530.0	804.2	158.0	1.108e-3	221.50	178.14	—

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.4	1.244e-3	248.78	200.08	—
6	-400.0	-530.0	804.2	1337.1	-2.005e-3	-400.00	-321.70	—
7	-200.0	-530.0	804.2	1285.4	-1.869e-3	-373.74	-300.58	—
8	0.0	-530.0	804.2	1233.7	-1.732e-3	-346.46	-278.64	—
9	200.0	-530.0	804.2	1182.1	-1.596e-3	-319.18	-256.70	—
10	400.0	-530.0	804.2	1130.4	-1.460e-3	-291.91	-234.76	—
11	-680.0	300.0	804.2	607.6	-7.922e-5	-15.84	-12.74	—
12	-680.0	100.0	804.2	800.8	-5.893e-4	-117.86	-94.79	—
13	-680.0	-100.0	804.2	994.0	-1.099e-3	-219.88	-176.84	—
14	-680.0	-300.0	804.2	1187.2	-1.609e-3	-321.89	-258.88	—
15	680.0	300.0	804.2	256.2	8.483e-4	169.65	136.44	—
16	680.0	100.0	804.2	449.4	3.382e-4	67.64	54.40	—
17	680.0	-100.0	804.2	642.7	-1.719e-4	-34.38	-27.65	—
18	680.0	-300.0	804.2	835.9	-6.820e-4	-136.40	-109.70	—

Solver and admissibility evidence

Strain plane ϵ_0	-3.8060e-4	Curvature k_x	2.5504e-6
Curvature k_y	6.8197e-7	Neutral-axis line	165.0295°
Compression depth c	577.545 mm	Iterations	6
Relative residual	1.419e-16	Max concrete compression	0.001525
Concrete limit	0.003300	Max steel tension	0.002005
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	0.8774		